Contract pricing model: technical design

Data model, contract selection, resolver and price calculation · for the development team

Status: Draft v4 proposal for Greg Smith's review. Supersedes v3 where they differ.
Based on: Greg Smith's *Contract pricing model: design summary* (v3) and *Identification of the Billable Party* notes, revised after the 5 October 2026 review with Donald Fraser
Companion document: *How procedures are priced and who pays* gives the same rules in business language for AA directors, anaesthetists and the office

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Pricing Model
Who pays and how much

1. Purpose

This model manages the pricing inputs for anaesthetic procedures and decides who pays. It rests on one separation: the **booking procedure** records what the anaesthetist did, and the **contract** defines how it is paid for. They are kept apart because the same procedure can be paid in different ways for different surgeons, hospitals, insurers and anaesthetists.

The RVG is an anatomically grouped list of recommended Base units, used in the calculation of the fee. The system should support the ability to associate a Procedure (in a Booking) with an RVG Base value, using a lookup capability. The design recognises that the RVG Base Unit allocation is a light-weight task that Anaesthetists may prefer to manage at the point in time of the procedure.

The model meets it with three ideas:

- **Defaults live once.** The RVG Groups table holds the published Base unit values. Booking Procedures start from them and override where needed.
- **Contracts are sparse overrides.** A contract line populates only the fields it sets. Everything else shows through from data below.
- **One path for every booking procedure.** The "No contract" scenario is modelled as a Default contract, so the price resolver and the billing engine never branch on whether a contract exists, they always follow the same logic. It's cleaner.

This document describes a logical model. Data types, keys and indexes are left to the physical design.

2. Terms

Term	Meaning	Business wording
Booking	One patient's appointment on a List. Holds one or more surgical procedures.	Booking (the term "Card" has been retired)
Booking Procedure	A Procedure contained in a booking, with its associated contract, BTM units and price. Replaces v3's "episode". A booking may contain multiple procedures.	Procedure
Time Entry	The anaesthetist's recording of BTM units, modifier notes, and any specific price or discount overrides against a booking procedure in the booking. Replaces v3's "timesheet".	Filling in the card
Contract Holder	The party a contract belongs to: an insurer, hospital, surgeon's rooms, or an anaesthetist (for their own price lists).	Contract holder
Contract	A dated set of pricing terms belonging to one Contract Holder. An arrangement not intended to imply a legal contract.	Contract
Contract Line	The procedure pricing details for one line within a contract.	A line of the Contract
Default Contract	The single contract flagged *is_default=true*. AA is the Contract Holder, with *is_billable_party=false*. This is how the system internally handles RVG based procedure pricing.	"No contract (RVG)"
Billable Party (BP)	Whoever receives the invoice: the Contract Holder's billable party record, or the payer/guardian named on the booking.	Who pays
Payer	The name and details of the Billable Party captured on the booking. Defaults to the patient and editable (for example to a nominated guardian).	Patient or guardian
BTM	Base, Time and Modifier units.	BTM
Resolver	The function that uses the information in a Booking Procedure to calculate and return the Booking Procedure price. Calculated as at the procedure date.	Not used
Price source	Which rule produced the final price (section 9).	Not used (?)

3. Data model

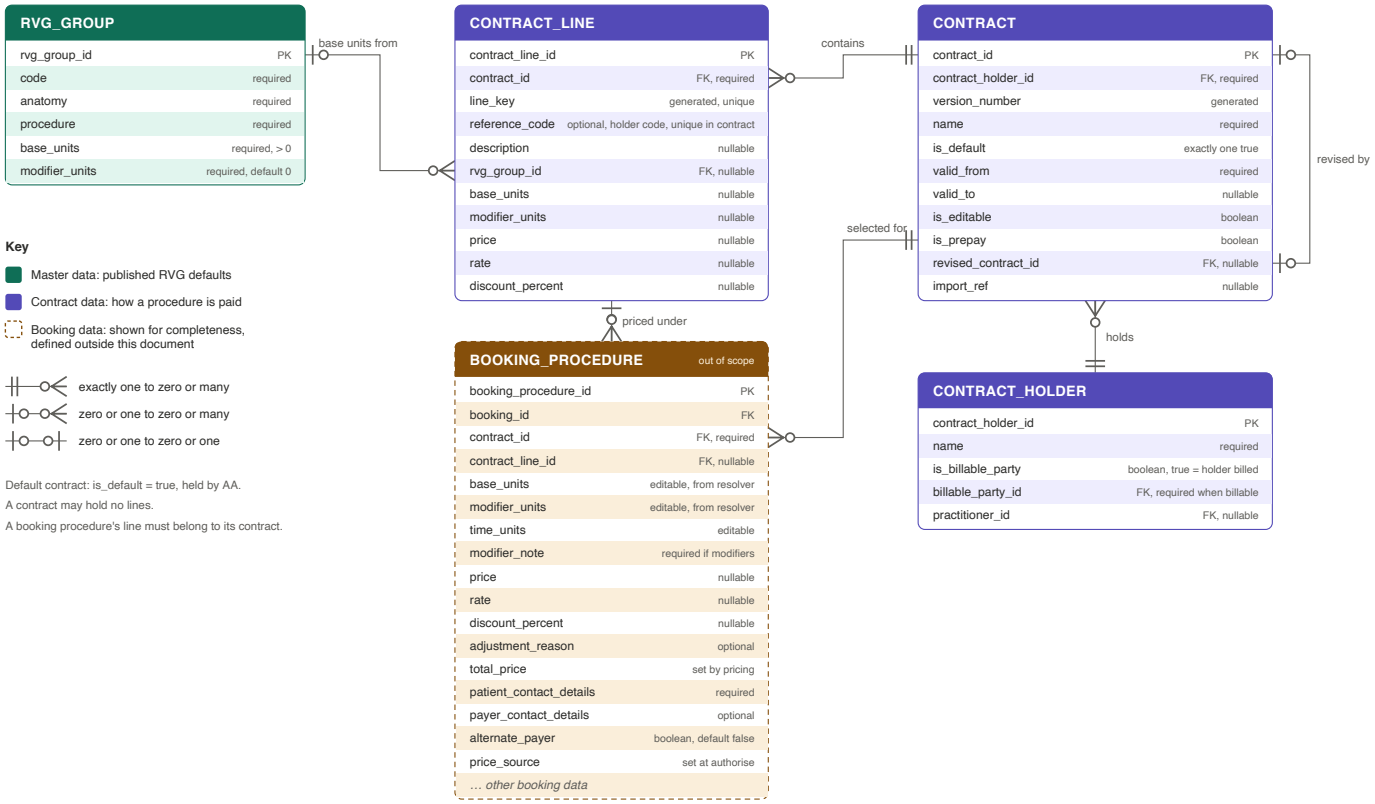


Figure 1. Revised contract pricing model. Precedence rises from the master data (green) to the contract overrides (purple). BOOKING_PROCEDURE shows only the fields pricing needs.

4. Table Definitions

RVG_GROUP (master data)

The published RVG group, as listed in the NZSA guide. AA may add its own groups.

Field	Rule	Notes
rvg_group_id	PK	Surrogate key. RVG codes are not unique enough to be keys.
code	Required	As published
anatomy	Required	As published.
procedure	Required	As published
base_units	Required, > 0	The bottom layer of the resolver, so base units always resolve.
modifier_units	Required, default 0	The bottom layer of the resolver, default to 0. This is how the system populates the default modifier value.

Design Revision: The Booking Procedure contains the default invoice description (not shown). So Booking Procedure will require 2 fields: booking_description and invoice_description, to accommodate the case where a default description is replaced by a contract line description.

CONTRACT_HOLDER

The party a contract belongs to. Two new behaviour fields replace v3's assumption that the holder is always billed.

Field	Rule	Notes
contract_holder_id	PK	
Name	Required	For example "Southern Cross", "Merivale Plastics", "Dr B. Smith prepaid list".
is_billable_party	boolean	True when AA invoices the holder (insurer, hospital, etc). In all other cases (false) AA invoices the patient/guardian (eg surgeon fixed-fee arrangements, anaesthetist lists, default contract).
billable_party_id	FK, nullable	Links to the existing billable party record. A billable party may be referenced by multiple contract holders. (This would be unusual).
practitioner_id	FK, nullable	Set where Contracts are used to specify anaesthetist specific "contract", typically used to setting prepay price lists. Can also be used to indicate when the contract is with a surgeon (this is similar to the anaesthetist) but is usually not prepay. Used to ensure that these contracts are only selectable when managing bookings for this anaesthetist and/or surgeon

CONTRACT

A dated set of terms. A Contract Holder can have many contracts. A Contract is kept as a series of dated versions: every change, whether an alteration or a replacement schedule, creates a new version (section 12).

Field	Rule	Notes
contract_id	PK	
contract_holder_id	FK required	
version_number	Generated	A system generated sequential number, applied as each Contract (for this Contract Holder) is created.
name	Required	Display name. The default contract shows as “No contract (RVG)”.
is_default	Exactly one true	The default contract is found by this flag, never by its name.
valid_from, valid_to	valid_from is mandatory, valid_to is nullable	Validity sits here, not on lines (changed from v3). Null valid_to means open-ended.
is_editable	boolean	A flag to indicate if this contract’s lines pricing is overridable (by the anaesthetist). Admins can always override.
is_prepay	boolean	A flag to indicate if prepayment is required for these contract lines.
revised_contract_id	FK, nullable	Forward link from a version to the version that replaced it. Set on submit of the new version.
import_ref	Nullable	Points to the upload or edit session, and its audit records, that created this version.

Deleted: A Contract is loaded from a schedule, and is a full replacement of the prior contract

Deleted: Forward links a price-review version to the contract that replaced it. (TBC)

Deleted: Points to the import log entry that created this version. (TBC)

CONTRACT_LINE

A contract’s terms for an (undefined) group of procedures (e.g Cataracts), which references one RVG group for unpriced work, or has a fixed price/rate/discount specified.

Field	Rule	Notes
contract_line_id	PK	
contract_id	FK, required	
line_key	Generated	System-generated identity of the line: unique and never reused. Kept when a line is carried into a new version, and used to relink booking procedures across versions. Included in an exported template.
reference_code	Optional	The contract holder’s own code for the line (hospitals supply their own). Unique within a contract when present. Used to match rows in a holder’s file, and to carry a line’s line_key forward in a Replace upload. See section 12.
description	Nullable	The description. If null use Booking Procedure description on invoice
rvg_group_id	FK, nullable	This Contract Line references (uses) this RVG Line’s Base units
base_units, modifier_units	Nullable	Override Base or Modifier units. 0 is a value that masks the default values.
price	Nullable	Either a fixed price, or prepay amount, for the procedure. BTM data is ignored.
rate	Nullable	An override unit rate in place of the anaesthetist’s default rate.
discount_percent	Nullable	A contract-level percentage discount.

Deleted: Required

Deleted: The Contract Line’s unique code, or equivalent and is used to match and update versions of contract lines. When loading schedules, this code needs to be unique within a single contract, as the contract lines schedule may be loaded more than once in different contract contexts.

BOOKING_PROCEDURE

Note that the Price and Discount %ge data may be made not-editable by the Contract. This is to prevent an anaesthetist overriding a fixed price contract, where this is required.

Field	Rule	Set by	Notes
booking_procedure_id	PK		
contract_id	FK, nullable	Office can setup; anaesthetist may change	
contract_line_id	FK, nullable	Office can setup; anaesthetist may change	One contract line per booking procedure. Initially set by Admins. Must be not null for billing .
base_units	editable	Default from RVG, editable by anaesthetist	Base inherited from RVG_Group.
time_units	editable	Anaesthetist at time entry	Not required if Price is specified.
modifier_note	Editable	Anaesthetist	Required when modifier units are claimed. Insurers ask for it.
price	May be editable	Anaesthetist, for editable contracts only	A multi-use field to hold either a fixed price, a prepayment amount or a price override. Null otherwise, may be zero.
discount_percent	May be editable	Anaesthetist, when editable	Percentage only. A %ge discount to be deducted from the total price. Details shown on invoice.
rate	Not editable	Contract	Allows for rate contracts
adjustment_reason	May be editable	Anaesthetist	Optional
total_price	Not editable	Pricing routine	Somewhere to keep the billed amount
patient_contact_details	Required	Multiple fields	Also the default billable party
payer_contact_details	Optional	Admin team	May be null
alternate_payer	boolean, required.	Admin team	Defaults to false
price_source	Required	Billing engine at authorisation	Records the way in which the price and discount were derived. Section 9. (?Price Source TBC)
insurer	May be blank	Admin, Anaesthetist	Txt, name of insurance provider
insurance_acceptance	May be blank	Admin, Anaesthetist	Reference code when insurance claim has been accepted
... other booking data elements			

5. Contract Behaviour

Four flags drive who is billed, and how, and what the anaesthetist can change.

Contract_Holder.is_billable_party flag
If True, indicates the billable party is the Contract Holder, otherwise the Billable Party is always the Patient/Guardian.

Contract.is_editable flag
If True, indicates if the anaesthetist may change the values of any of the contract lines. For Hospital fixed price contracts this is typically set to False to prevent the anaesthetist overriding the price and discount fields for Fixed Price contracts. Conversely, for procedures using RVG values, it is set to True.

Contract.is_prepay flag
If True, triggers an automated prepay process (letter, invoices etc)

Contract_Holder.practitioner_id not null
If set, indicates that this procedure uses an anaesthetist specific configuration (usually fixed-price, prepay), or a surgeon specific configuration (fixed-price).

This is designed to support prepay requirements for anaesthetists. The typical scenario configuration would be: The Billable Party is the patient, the anaesthetist is nominated, the arrangement is pre-pay, and the amount is editable by the anaesthetist.

How the cases map

Typical Arrangement examples	is_billable_party	practitioner_id	is_editable	is_prepay
Default Hospital contract (using RVG)	true	null	true	false
Hospital contract, fixed priced/rate	true	null	false	false
Default Contract (uses RVG), patient pays	false	null	true	false
Anaesthetist fixed price (prepay) list	false	set	true	true
Surgeon's fixed-fee arrangement	false	set	false	false

Various other combinations can be used to achieve specific billing outcomes.

The office can override any pricing at any time. That admin override is the only way to change a non-editable price without changing the contract.

6. Contract selection

1. **Billable Party identification, at booking setup.**
   The admin reviews the booking to ascertain the hospital, surgeon, and procedure description.
   The admin then identifies the relevant Billable Party by selecting a Contract Holder and then Contract (may be Default RVG contract).
   The default contract should always appear first in any list.
   It is followed by every contract that is valid on the procedure date, and fits the booking's context: the hospital, the surgeon or rooms, and anaesthetist-specific lists etc.
2. **Office, at booking setup.**
   a. Where the Default Contract (RVG) is selected, the admin may select from either the most relevant entry from the RVG to set the default Base Units or may select from a pre-loaded list of routine procedures (which may be pre-loaded by the user as Contract Lines in the Default contract).
   The admin may also leave this blank, for the anaesthetist to assign at a later time.
   b. Where a non-Default Contract has been selected, the admin selects the most relevant Contract Line, which may specify any of the pre-defined values for: RVG Base Units, Price, Rate, or Discount.
   The admin may also leave this blank, for the anaesthetist to assign at a later time.
3. **Anaesthetist, at time entry.**
   At any time before the list is submitted, the anaesthetist may: change the procedure, the contract, the RVG code, Price, Rate or Discount (where permitted). No reason is needed. The change is audited and flagged at office review.

4. **On any change**
   The resolver re-runs, and the pricing variables (Base, Price, Rate, %ge) refresh.

7. The Resolver

The Resolver calculates the pricing for the procedure. It checks for Contract validity and uses the values in the Booking Procedure.

The need to check the Contract validity is that various Contract arrangements (fixed prices) may have changed in the time between when the Booking was first made, and when the Procedure was carried out.

The resolver checks that the Contract date is valid and, if necessary, refreshes any data from the revised Contract Line.

Inputs

The Procedure data within a Booking. (Base, Time, Modifiers, Price, Rate, %ge)

Pricing Sequence, highest precedence first

1. The Price (if entered) overrides any other calculation.
2. The Rate, if entered, overrides the Anaesthetist's personal rate.
3. The BTM values multiplied by the Rate (the Anaesthetist's own or the override value)
4. The discount percentage is applied, if specified

Any discounts calculated are shown on a separate invoice line.

Output

The resolved values, plus the layer each one came from. Both go into the pricing snapshot.

Treatment of nulls

- Null means "not specified here", and resolution looks through to the next layer. A value, including 0, stops it.
- An explicit 0 above the RVG level is only needed to switch off a non-zero value beneath. That is rare, but some contracts could zero modifiers, so the distinction matters.
- In editing screens, each overridable field shows the inherited value as a greyed placeholder, with an explicit choice between **inherit** (stored as null) and **override** (an override with no value stores 0).
- In import spreadsheets, a blank cell means inherit and an explicit 0 means zero.

8. Time entry

- The resolved base and modifier units are shown as starting values. The anaesthetist enters time units and may change base and modifier units. Out-of-range values warn the office. They are never blocked.
- Modifiers are itemised, with a required `modifier_note` when any are claimed. Whether modifiers are ever pre-filled is open (section 15).
- The price field and the discount follow the contract editing behaviour table in section 5.
- The record keeps the values the anaesthetist recorded. Starting values can always be recomputed by the resolver, and they are captured in the snapshot at authorisation.

9. Price calculation

The billing engine prices each Procedure in a Booking on its own when the list is authorised.

Validation first

The Time Entry process prohibits entry of price or discount when a contract is not editable

The Time entry process rejects Submission/Update where:

- No Price resolves
- The selected contract is not valid on the procedure date – if not resolved, use original data, and flag a warning

However, a method is required for the anaesthetist to forward an incomplete Booking Procedure to the Admin team for resolution, where any issues arise.

Precedence/Sequence

Step	Condition	Price	price_source
0	The office overrides the price at post-procedure review	The office price. Post procedure. (Alters *Booking_Procedure.Price*)	`OFFICE_OVERRIDE`
1	`Any of Price, Rate or discount` are altered during Time Entry event	The anaesthetist overrides the price.	`ANAESTHETIST_PRICE`
2	`A contract price` resolves	Either a fixed-price, or price not edited	`CONTRACT_PRICE`
3	RVG Pricing	(B + T + M) × rate × (1-discount_percent)/100, where rate is the resolved contract `rate` if any, else the anaesthetist's own unit value	`CALCULATED`

- A price of 0 produces a no-charge invoice. BTM is still recorded, if entered.
- The multi-procedure rules (for example base units on the primary procedure only, allocation of Modifiers) are applied to adjust the units of `CALCULATED` procedures before step 3.

Snapshot

At authorisation, each booking procedure stores the procedure, the contract (each version is its own row, so the id is enough), the resolved values and their layers, the recorded BTM, the rate and discount used, the price, the `price_source`, and the billable party or payer details. Later changes to the RVG, procedures or contracts never alter historical billing, and anaesthetist-nominated prices are easy to report on.

10. Billable party

Section 5 covers the configurations that determine who is the billable party. Below is a high-level outline of how the logic works in relation to Billable Party.

Selected contract	Invoice goes to
Default RVG contract	The patient by default, editable to be another person or a guardian
Contract Holder.is_billable_party = true	The Contract Holder itself, or nominated billable party
Contract Holder.is_billable_party = false	The patient, or other nominated payer

- The surgeon's booking, or the hospital update, may indicate insurance status (for example SXAP). If the booking carries such an indication, a patient-billed contract with an "insurer pays" indication raises a warning

at review. It is not a block, because pre-approval may fall through on the day and the anaesthetist then switches to the default contract and bills the patient. TBC

- AA acts as agent for independent anaesthetists, so the system permissively accepts the anaesthetist's choice of contract and used on office review to catch mistakes.

11. Prepayment

Prepayment arrangements use the contract pricing model to capture anaesthetists prepayment price specifications. The points that touch this model:

- **Trigger.** The admin managing the booking process, selects a Contract that specifies *is_prepay = true*.
- **Amount:** The Contract Line will always specify an (estimated) fixed price.
- **Settlement.** The prepaid amount behaves as an editable fixed price.
  - If the price (displayed on the booking) is NOT overridden, the final price equals the prepaid amount and nothing more is billed.
  - If the anaesthetist raises the price, the difference is billed as an additional invoice.
  - If the anaesthetist lowers the price (by altering the price or applying a discount) a refund process is invoked for the difference.

12. Updates, history and data loading

Contracts change in two ways: a correction or mid-year update to some of their lines, and a periodic renewal of the whole schedule. Both are handled on one screen for a single contract and follow one rule: **every change creates a new contract version from a date the office chooses.** Nothing is backdated and nothing is deleted.

Maintaining a contract

- The office maintains a contract on a header and lines screen. The header holds the name, dates and flags; the lines sit below it. Both can be edited on screen, and lines can also be loaded from a schedule upload in the AA template.
- Edits and uploads are not applied straight away. They are staged and shown on a review dashboard for that contract. The office reviews them, then submits. On submit the system creates the new version and writes audit records of what changed.
- Screen edits and uploads follow the same rules and can be combined in one review.

Two modes: Alter and Replace

	Alter	Replace
Used for	Correcting an error. Mid-term adds and updates to a subset of lines (some hospitals send these regularly).	Periodic renewal, or any complete new schedule.
New version starts with	A copy of every line in the current version.	No lines.
Uploaded rows	Applied to the copy according to each row's Operation (Add, Alter, Remove).	Form the complete new schedule. The Operation column is ignored.
Lines not in the upload	Carried forward unchanged.	Not carried forward. They remain on the previous version only.

Effective dates

- Validity stays on the contract header, not on lines. Every submit, in either mode, creates a new version.
- **Effective From starts blank and must be entered before submit.** There is deliberately no default: a default of today or tomorrow would nearly always be wrong.

Deleted: Contract versions¶
A Contract should be able to be Replaced or Altered. ¶
An update strategy could incorporate a manually managed Contract, supported by a list upload of Contract_lines.¶
Date Validity is held at CONTRACT level. An update is effected by uploading of a set of contract lines. It replaces, or is merged with, the existing contract lines. ¶
The Contract Line is cross-referenced to bookings contract_line.line_reference_code¶
A preview should list new, changed and removed lines for approval before anything is applied. `import_ref` points to the import log. Ideally, a review will highlight any removed lines that still have live Booking Procedures.¶
Contract holds a forward reference to a newer Contract revision, to allow date validity checking and correction from the Booking Procedure. ↵
If a forward reference to *Contract.revised_contract_id* and *Contract Line.reference_code* cannot be found, use the original value in Contract Line.↵
This feature could be used to revalidate all existing lines at the time a Contract revision is applied. This would obviate the need for Bookings to revalidate details.¶
The **delta** upload feature is not available¶
A row is the complete line, not a patch: a blank cell inherits and 0 means zero.¶

- Changes are never backdated. Effective From cannot be in the past.
- On submit, the previous version's `valid_to` is set to the day before the new Effective From, and the new version is open-ended (`valid_to` is null). Consecutive versions never overlap and never leave a gap.
- The procedure date decides which version prices a booking procedure. A procedure performed before the Effective From is priced from the earlier version, even if it is invoiced later.
- The management of prepay and surgeon fixed price contracts is not fully resolved in this scenario
- A contract ends without a successor only when the office enters a `valid_to` on its current version.
- `version_number` is generated. The previous version's `revised_contract_id` points to the new version, and `import_ref` records the upload or edit session that created it.

Example. Hospital X's contract version 3 has been valid from 1 April with no end date. On 20 October the hospital sends updates to six lines, effective 1 November. The office uploads the six rows in Alter mode, enters 1 November as Effective From, reviews and submits. Version 4 is created with all version 3 lines plus the six changes, valid from 1 November. Version 3 now ends on 31 October. A procedure performed on 25 October is priced from version 3; a procedure booked for 3 November is relinked to the matching version 4 line.

Future-dated versions and operator error

- A contract may have only one future-dated version at a time.
- Until its Effective From is reached, nothing has been priced from it, so it can be corrected freely through the same review and submit:
  - amend its lines or header;
  - change its Effective From (the current version's end date moves with it);
  - withdraw it. The current version becomes open-ended again, and any booking procedures already relinked return to the matching current lines.
- Once a version is in effect, an error is corrected with a new version from a chosen date. Any affected procedures in between can be adjusted with the office price override.

Line identity: two codes

- Each contract line carries two codes:
  - `line_key` is generated by the system, unique and never reused. When Alter copies a line into a new version, the line keeps its `line_key`. It is how booking procedures are relinked across versions, and it is included in an exported template.
  - `reference_code` is the contract holder's own code for the line. Hospitals supply their own codes. It is optional, and unique within a contract when present. It lets a hospital's file be matched without anyone looking up system codes.
- Upload rows are matched on `line_key` when it is present, otherwise on `reference_code`.

The upload template

- One row per contract line, in the AA template. The mode (Alter or Replace) is chosen on the upload screen, not in the file.
- A row is the complete line, not a patch. A blank cell is null, so the value shows through from the data below it (the RVG group). A filled value, including 0, overrides everything below it. An altered row therefore replaces the whole line: a row missing values will clear them.
- In Alter mode, an Operation column says what each row does:

Operation	Matching line in the current version	Result in the new version
Add	Must not exist	Line added. The system assigns its `line_key`.
Alter	Must exist	Line replaced by the row.
Remove	Must exist	Line not carried forward. Booking procedures on or after the Effective From that used it are flagged.

Operation	Matching line in the current version	Result in the new version
Blank	Any	Row ignored; the line is carried forward unchanged. A warning is shown if the row differs from the current line.

- The Operation is checked, not guessed. Add on a code that already exists, or Alter or Remove on a code that does not, is an error. This catches mistyped codes, which would otherwise create near-duplicate lines.
- In Replace mode the Operation column is ignored, and a row marked Remove raises a warning because it suggests the wrong mode. New rows are matched to the previous version by `reference_code` so that they keep their `line_key`; rows with no match get new keys.
- Exporting the current contract to the template, with its `line_key` values, is the easiest way to prepare an Alter upload. It is recommended for the MVP.

Review dashboard

- Before submit, the dashboard shows:
  - header changes, before and after, and the Effective From;
  - line counts: added, altered, removed and unchanged;
  - for each altered line, the fields that change, before and after, with any value that becomes blank highlighted;
  - errors, which block submit: duplicate codes in the file, Operation mismatches, an unknown RVG group, invalid values, a missing or past Effective From, or a second future-dated version;
  - warnings, which do not block submit: booking procedures affected by altered or removed lines and, for Replace, previous lines with no match in the new schedule.
- Submit applies everything or nothing.

Effect on bookings

- On submit, un-invoiced booking procedures dated on or after the Effective From are relinked from the previous version to the new version's line with the same `line_key`.
- Where there is no matching line (it was removed, or is not in a Replace schedule), the booking procedure is listed on the dashboard for the office to resolve.
- Procedures dated before the Effective From are untouched. Invoiced procedures are also protected by the snapshot.

Data loading

- All master data is prepared by AA as spreadsheets in AA-defined templates: RVG groups, procedures, contract holders, contracts and lines, plus hospitals, surgeons and rooms, and anaesthetists.
- Loading must be repeatable, so test environments can be wiped and reloaded.
- The loading mechanism is open. The suggested MVP baseline is a developer-run import, then office screens to create, edit and retire records. Ongoing contract maintenance then uses the in-app upload and review dashboard described above.

Deleted: An in-app upload with preview, and exporting a contract to the template for its holder to update, are later options....

History

Creates, updates and retirements of RVG groups, procedures, contract holders and contracts are audited (who, what, when). That replaces v3's suggestion of modelling the RVG as the lowest contract to keep its history. Each contract submit also records the lines added, altered and removed, with before and after values, linked through import_ref. Billing history is covered by the snapshot.

Deleted: Billing history is covered by the snapshot.

13. Integrity rules

- Multiple Contract Lines (in a Contract) may reference the same RVG line

- No overlap and no gap between consecutive versions of a contract. At most one future-dated version per contract. A new version's Effective From is never in the past.
- `line_key` is unique and never reused. `reference_code`, when present, is unique within a contract version.
- Contract has valid date, otherwise substitute values current at date of procedure
- Exactly one contract has *`is_default`* = *true*. *Contract.valid_to_date* = null
- `base_units and modifier_units` are optional for the user at time entry
- *`is_billable_party`= true* requires `billable_party_id`.
- The time entry function rejects a price or discount change on a contract that is not editable. The underlying fixed price contract may have any combination of the pricing variables.

Deleted: No overlapping validity between a contract and its previous or next version.

14. Routine/Regular Procedures

Some procedures are performed on a regular basis. The intent (in the context of this design) is to include the ability to load a list of these as Contract Lines in the Default Contract. This allows an admin or anaesthetist to choose this pre-defined procedure rather than just specifying the RVG Base code.

This is the same mechanism as choosing a Contract Line (for example) from a Fixed Price contract.

In situations where a Contract Line is selected, the Description from that Contract Line is used as the description on the invoice, rather than the description from the Booking Procedure details.

Out of scope for this document

The booking and list lifecycle, invoice generation, the import log, and the billable party record are referenced but not modelled here. No nesting of individual contract lines is provided. Each stands alone.

15. Changes from v3

v3	v4	Why
“Episode”, “timesheet”, “card”	Booking procedure, time entry	One vocabulary across the documents and requirements.
`valid_from`/`valid_to` on contract lines	Validity on CONTRACT only. Price reviews create versions.	All lines in a contract share its dates (Greg, 5 Oct).
Default contract reached through its lines (TBC)	Offered for every procedure by rule.	
Holder always billed via `billable_party_id`	Payer captured on the booking. See Section 5	Surgeon fixed-fee arrangements set the price, but the patient pays.
“Timesheet price overrides all other logic”	True, but may not be editable	Fixed pricing contracts cannot be changed by the anaesthetist.
Timesheet price valid only with no contract	See Section 5	Anaesthetists keep discretion over their own pricing, including prepaid amounts.
Prepaid difference becomes a balance due or a refund	An increase is invoiced. A refund may be issued	Agreed in the 5 Oct review. AA may refund prepayments.
“Fixed base units” contract parameter	“Base units” override	They are starting values, not fixed.
Practice-wide self-pay price via default contract lines	Removed	AA cannot set a common price for its members.
Billing issues a receipt, not an invoice, when prepaid	Removed	Not confirmed. Handled by the prepayment specification.
`modifiers_excluded` flag; per-modifier child table	Removed	Out of scope. The anaesthetist itemises modifiers in the note.
RVG history via a lowest-precedence contract	Audit records on update	Simpler, and sufficient.
Full upload removes missing lines	Alter (merge by Operation) or Replace. Every change creates a new dated version.	Hospitals send regular adds and updates. Nothing is backdated or deleted.
Override units “with an explanation”	Note required for modifiers. None for a contract change.	Insurers ask for modifier explanations. Contract changes are flagged at review instead.

Deleted: Closes them with the old version

Deleted: Nothing is deleted.